

NFPA 780—DETAILED LIGHTNING RISK ASSESSMENT (LRA)

Both a detailed and a simplified LRA were conducted for the site. The following pages show the detailed 780 LRA that was conducted for the site.

Lightning Collection Areas

This spreadsheet shows the calculations of lightning collection areas from the NFPA 780 Detailed LRA that was conducted for the site. Values for AD and AM were calculated with graphical Dehn CAD software shown on the subsequent pages.

Lightning Collection Areas				
Equivalent Collection Area of Structure				
Inputs				
Height of structure	H =	7.92 m	Electrolyzer Building	
Width of structure	W =	53.34 m		
Length of structure	L =	22.86 m		
Calculation				
$A_D = LW + 6H(L + W) + 9\pi H^2$	$A_D =$	6613.9 m ²	Area within 3x height of structures.	
Note: Calculation is only valid for rectangular or near-rectangular structures.				
Collection Area of Strikes Near Structure				
Inputs				
Width of structure	W =	53.34 m	See dimentions above.	
Length of structure	L =	22.86 m		
Calculation				
$A_M = LW + 500(L + W) + \pi(250)^2$	$A_M =$	235,669 m ²	Area within 250 m of structure	
Note: Calculation is only valid for rectangular or near-rectangular structures.				
Equivalent Collection Area of Adjacent Structure				
Inputs				
Height of adjacent structure	H _J =		m	Linde Plant
Width of adjacent structure	W _J =		m	
Length of adjacent structure	L _J =		m	
Calculation				
$A_{DJ} = LW + 6H(L + W) + 9\pi H^2$	$A_{DJ} =$	125856	m ²	DEHN Software calculation Area within 3x height of adjacent structure.
Note: Calculation is only valid for rectangular or near-rectangular structures.				
Equivalent Collection Area of Incoming Service				
Inputs				
Length of incoming service line	L _L =	240 m	Use 1000 m, if greater than 1 km.	
Calculation				
$A_L = 40 * L_L$	$A_L =$	9,600	m ²	
Collection Area of Strikes Near Incoming Service				
Inputs				
Length of incoming service line	L _L =	240 m	See above.	
Calculation				
$A_I = 4000 * L_L$	$A_I =$	960,000	m ²	

ANNUAL THREAT OCCURRENCE

Annual Threat Occurrence			
Direct Strike to Structure			
<i>Inputs</i>			
Lightning flash density	$N_G =$	1.19 flashes/km ² /year	Calculated in another section of this report.
Equivalent collection area	$A_D =$	6,614 m ²	Area within 3x height of structure
Location coefficient	$C_D =$	0.5 (unitless)	Structure surrounded by structures of equal or lesser height within a distance of 3H
<i>Calculation</i>			
$N_D = (N_G)(A_D)(C_D)(10e-6)$	$N_D =$	0.004 events/year	Approximately, one event every 250 years.
Strikes Near Structure			
<i>Inputs</i>			
Lightning flash density	$N_G =$	1.19 flashes/km ² /year	
Collection area near structure	$A_M =$	235,669 m ²	Area within 250 m of structure
Equivalent collection area of structure	$A_D =$	6,614 m ²	
Location coefficient	$C_D =$	0.5 (unitless)	
<i>Calculation</i>			
$N_M = (N_G)(A_M - A_D)(C_D)(10e-6)$	$N_M =$	0.14 events/year	Approximately, one event every 7 years.
Direct Strikes to Incoming Service			
<i>Inputs</i>			
Lightning flash density	$N_G =$	1.19 flashes/km ² /year	
Equivalent collection area of incoming service	$A_L =$	9,600 m ²	
Environmental coefficient	$C_E =$	1.0 (unitless)	Rural
Transformer coefficient	$C_T =$	0.2 (unitless)	Transformer between site and incoming service
<i>Calculation</i>			
$N_L = (N_G)(A_L)(C_E)(C_T)(10e-6)$	$N_L =$	0.0023 events/year	Approximately, one event every 436 years.
Direct Strikes to Adjacent Structure			
<i>Inputs</i>			
Lightning flash density	$N_G =$	1.19 flashes/km ² /year	
Equivalent collection area of adjacent structure	$A_{DJ} =$	125,856 m ²	Area within 3x height of adjacent structure
Environmental coefficient	$C_D =$	0.5 (unitless)	Structure surrounded by structures or trees of equal or lesser height within a distance of 3H
Transformer coefficient	$C_T =$	0.2 (unitless)	Transformer between site and adjacent structure
<i>Calculation</i>			
$N_{DJ} = (N_G)(A_{DJ})(C_D)(C_T)(10e-6)$	$N_{DJ} =$	0.02 events/year	Approximately, one event every 67 years.
Strikes Near an Incoming Service			
<i>Inputs</i>			
Lightning flash density	$N_G =$	1.19 flashes/km ² /year	
Collection area near incoming service	$A_I =$	960,000 m ²	Underground cables have are assumed to have a lightning collection area of zero.
Service environmental coefficient	$C_E =$	1.0 (unitless)	Rural
Transformer coefficient	$C_T =$	0.2 (unitless)	Transformer between site and service line
<i>Calculation</i>			
$N_I = (N_G)(A_I)(C_E)(C_T)(10e-6)$	$N_I =$	0.2 events/year	Approximately, one event every 4 years.

PROBABILITY OF DAMAGE

Probability of Damage			
Injury Due to Direct Strike			
See Table L.6.7.2	$P_A =$	0.000001 (probability)	Structural steel frame is used as the down conductor system.
Physical Damage Due to a Direct Strike			
See Table L.6.7.3	$P_B =$	0.001 (probability)	Structure with a metal roof meeting the requirements of 4.6.1.4, with continuous metal serving as a natural down conductor system with bonding and grounding in accordance with NFPA 780.
Failure of Internal Systems Due to a Direct Strike			
See Table L.6.7.4	$P_C =$	1.00 (probability)	No SPD protection.
Failure of Internal Systems Due to a Strike Near Structure			
<i>Inputs</i>			
Spacing between steel columns	$W_M =$	5.8 m	Approximately 19 feet between steel columns.
Lowest withstand voltage in system	$U_W =$	1.5 kV	For structures containing computer equip, use 1.5 kV.
Factor for external shielding	$K_{S1} =$	0.69	If no shield, then use $KS1 = 0.12\omega$.
Factor for internal shielding	$K_{S2} =$	0.69	If no shield, then use $KS2 = 0.12\omega$.
Factor for internal wiring	$K_{S3} =$	1.0	Unshielded cable—no routing precaution to avoid loops
Factor for equipment withstand voltage	$K_{S4} =$	1.0	$KS4 = 1.5/UW$
Factor for internal systems protection	$K_S =$	0.48	$KS = (KS1)(KS2)(KS3)(KS4)$
<i>Results</i>			
See Table L.6.7.5	$P_M =$	1.0 (probability)	For values of $KS > 0.4$, then $PM = 1.0$
Injury Due to Strike to Incoming Service			
<i>Inputs</i>			
Lowest withstand voltage in system	$U_W =$	1.5 kV	See above.
Shield resistance of incoming service conductors	$R_S =$	$\leq 1 \Omega/\text{km}$	Shielded buried feeder
<i>Results</i>			
See Table L.6.7.7	$P_U =$	0.4 (probability)	Shielded buried line whose shield is bonded to grounding system
Physical Damage Due to Strike to Incoming Service			
Results	$P_V =$	0.40 (probability)	Where SPDs are provided, use the lower value between PC and PU.
Failure of Internal Systems Due to Strike to Incoming Service			
Results	$P_W =$	0.40 (probability)	
Failure of Internal Systems Due to Strike Near Incoming Service			
<i>Inputs</i>			
Lowest withstand voltage in system	$U_W =$	1.5 kV	See above.
<i>Results</i>			
See Table L.6.7.8	$P_Z =$	0.6 (probability)	For power and telecom lines.

LOSS FACTORS

Loss Factors				
Injury to Humans				
<i>Inputs</i>				
Typical mean loss of life factor	$L_T =$	0.001		<i>Persons inside and outside building</i>
Reduction factor for type of surface	$r_t =$	0.01		<i>Soil or concrete (Resistivity < 1 kΩ-m)</i>
<i>Calculation</i>				
$L_A = (L_T)(r_t)$	$L_A =$	0.00001	(loss factor)	
Physical Damage				
<i>Inputs</i>				
Typical mean property damage factor	$L_F =$	0.05		<i>Industrial, commercial, or school</i>
Reduction factor for provisions taken to reduce the consequences of fire	$r_p =$	1.0		<i>No provisions taken</i>
Reduction factor for risk of fire to structure	$r_t =$	0.10		<i>Explosion, Zone 1</i>
Hazard factor	$h_z =$	2		<i>Low level of panic (e.g., structures limited to two floors and the number of people not greater than 100)</i>
<i>Calculation</i>				
$L_B = (L_F)(r_p)(r_t)(h_z)$	$L_B =$	0.01	(loss factor)	
Failure of Internal Systems				
See Table 6.7.9	$L_o =$	0.000001	(loss factor)	<i>Industrial, commercial, or school</i>

RISK COMPONENTS

Risk Components			
Risk of Injury or Loss of Life from a Direct Strike to Structure			
<i>Inputs</i>			
Annual threat of direct strike to structure	$N_D =$	0.004 events/year	
Probability of injury or loss of life	$P_A =$	0.000001 (probability)	
Loss factor for injury or loss of life	$L_A =$	0.00001 (loss factor)	
<i>Calculation</i>			
$R_A = (N_D)(P_A)(L_A)$	$R_A =$	3.95E-14 (risk factor)	
Risk of Physical Damage from a Direct Strike to Structure			
<i>Inputs</i>			
Annual threat of direct strike to structure	$N_D =$	0.004 events/year	
Probability of physical damage	$P_B =$	0.001 (probability)	
Loss factor for physical damage	$L_B =$	0.01000 (loss factor)	
<i>Calculation</i>			
$R_B = (N_D)(P_B)(L_B)$	$R_B =$	3.95E-08 (risk factor)	
Risk of Failure of Internal Systems from a Direct Strike to Structure			
<i>Inputs</i>			
Annual threat of direct strike to structure	$N_D =$	0.004 events/year	
Probability of failure of int. systems	$P_C =$	1.0 (probability)	
Loss factor for failure of int. systems	$L_C =$	0.000001 (loss factor)	Note: $L_C = L_o$
<i>Calculation</i>			
$R_C = (N_D)(P_C)(L_C)$	$R_C =$	3.95E-09 (risk factor)	
Risk of Failure of Internal Systems from a Strike Near Structure			
<i>Inputs</i>			
Annual threat of strike near structure	$N_M =$	0.1 events/year	
Probability of failure of int. systems	$P_M =$	1.0 (probability)	
Loss factor for failure of int. systems	$L_M =$	0.000001 (loss factor)	Note: $L_M = L_o$
<i>Calculation</i>			
$R_M = (N_M)(P_M)(L_M)$	$R_M =$	1.37E-07 (risk factor)	
Risk of Injury or Loss of Life from a Direct Strike to Incoming Service			
<i>Inputs</i>			
Annual threat of strike incoming service	$N_L =$	0.0023 events/year	
Annual threat of strike to adj. structure	$N_{Dj} =$	0.0150 events/year	
Probability of injury or loss of life	$P_U =$	0.4 (probability)	
Loss factor for injury or loss of life	$L_U =$	0.00001 (loss factor)	Note: $L_U = L_A$
<i>Calculation</i>			
$R_U = (N_L + N_{Dj})(P_U)(L_U)$	$R_U =$	6.93E-08 (risk factor)	
Risk of Physical Damage from a Direct Strike to Incoming Service			
<i>Inputs</i>			
Annual threat of strike incoming service	$N_L =$	0.0023 events/year	
Annual threat of strike to adj. structure	$N_{Dj} =$	0.0150 events/year	
Probability of physical damage	$P_V =$	0.4 (probability)	
Loss factor for physical damage	$L_V =$	0.01 (loss factor)	Note: $L_V = L_B$
<i>Calculation</i>			
$R_V = (N_L + N_{Dj})(P_V)(L_V)$	$R_V =$	6.93E-05 (risk factor)	
Risk of Failure of Internal Systems from a Direct Strike to Incoming Service			
<i>Inputs</i>			
Annual threat of strike incoming service	$N_L =$	0.0023 events/year	
Annual threat of strike to adj. structure	$N_{Dj} =$	0.0150 events/year	
Probability of failure of int. systems	$P_W =$	0.4 (probability)	
Loss factor for failure of int. systems	$L_W =$	0.000001 (loss factor)	Note: $L_W = L_o$
<i>Calculation</i>			
$R_W = (N_L + N_{Dj})(P_W)(L_W)$	$R_W =$	6.93E-09 (risk factor)	
Risk of Failure of Internal Systems from a Strike Near Incoming Service			
<i>Inputs</i>			
Annual threat of strike near inc. service	$N_i =$	0.23 events/year	
Annual threat of strike incoming service	$N_L =$	0.0023 events/year	
Probability of failure of int. systems	$P_Z =$	0.6 (probability)	
Loss factor for failure of int. systems	$L_Z =$	0.000001 (loss factor)	Note: $L_Z = L_o$
<i>Calculation</i>			
$R_Z = (N_i + N_L)(P_Z)(L_Z)$	$R_Z =$	1.39E-07 (risk factor)	

ANNUAL RISK CALCULATIONS

Annual Risk Calculations			
Risk of Injury or Loss of Life			
<i>Inputs</i>			
Risk of injury or loss of life from a direct strike to structure	$R_A =$	3.95E-14	
Risk of physical damage due to a direct strike to structure	$R_B =$	3.95E-08	
Risk of injury or loss of life from a direct strike to incoming service	$R_U =$	6.93E-08	
Risk of physical damage due to a direct strike to incoming service	$R_V =$	6.93E-05	
<i>Calculation</i>			
$R_1 = R_A + R_B + R_U + R_V$	$R_1 =$	6.94E-05 (risk)	
Tolerable Risk for Loss of Life or Injury (Table L.6.2)	$R_{1T} =$	1.00E-05 (risk tolerated)	
<i>Note: Calculation not applicable for structures with life-critical electrical equipment, risk of explosion, or where failure of internal system immediately endangers life.</i>			
Risk of Loss of Service (Power, Phone, Water, etc.)			
<i>Inputs</i>			
Risk of physical damage due to a direct strike to structure	$R_B =$	3.95E-08	
Risk of failure of internal systems due to a direct strike to structure	$R_C =$	3.95E-09	
Risk of failure of internal systems due to a strike near structure	$R_M =$	1.37E-07	
Risk of physical damage due to a direct strike to incoming service	$R_V =$	6.93E-05	
Risk of failure of internal systems due to a direct strike to incoming service	$R_W =$	6.93E-09	
Risk of failure of internal systems due to a strike near incoming service	$R_Z =$	1.39E-07	
<i>Calculation</i>			
$R_2 = R_B + R_C + R_M + R_V + R_W + R_Z$	$R_2 =$	6.97E-05 (risk)	
Tolerable Risk for Loss of Service (Table L.6.2)	$R_{2T} =$	1.00E-03 (risk tolerated)	
Risk of Loss of Historical Significance			
<i>Inputs</i>			
Risk of physical damage due to a direct strike to structure	$R_B =$	n/a	
Risk of physical damage due to a direct strike to incoming service	$R_V =$	n/a	
<i>Calculation</i>			
$R_3 = R_B + R_V$. If not a historical building, $R_3 = 0$.	$R_3 =$	0.000E+00 (risk)	
Tolerable Risk for Loss of Historical Significance (Table L.6.2)	$R_{3T} =$	1.00E-03 (risk tolerated)	
Risk of an Economic Loss			
<i>Inputs</i>			
Risk of physical damage due to a direct strike to structure	$R_B =$	3.95E-08	
Risk of failure of internal systems due to a direct strike to structure	$R_C =$	3.95E-09	
Risk of failure of internal systems due to a strike near structure	$R_M =$	1.37E-07	
Risk of physical damage due to a direct strike to incoming service	$R_V =$	6.93E-05	
Risk of failure of internal systems due to a direct strike to incoming service	$R_W =$	6.93E-09	
Risk of failure of internal systems due to a strike near incoming service	$R_Z =$	1.39E-07	
<i>Calculation</i>			
$R_4 = R_B + R_C + R_M + R_V + R_W + R_Z$	$R_4 =$	1.89E-07 (risk)	
<i>Note: Calculation not applicable to structures where animals/livestock could be lost.</i>			

OVERALL ANNUAL RISK

The following spreadsheet lists the calculation results and Acceptable Risk Level from the NFPA 780 Detailed LRA that was conducted for the site.

Overall Annual Risk Calculation			
Overall Risk to the Structure			
<i>Inputs</i>			
Risk of injury or loss of life	R1 =	6.94E-05	
Risk of loss of service (power, phone, water, etc.)	R2 =	6.97E-05	
Risk of loss of historic significance	R3 =	0.00E+00	
Risk of an economic loss	R4 =	1.89E-07	
<i>Calculation</i>			
R = R1 + R2 + R3 + R4	R =	1.39E-04	(risk)

Overall Risk Results for NFPA 780 LRA

NFPA 780 Detailed LRA Conclusion

The NFPA identifies typical tolerable risk values for a lightning risk assessment. According to NFPA 780, the typical tolerable risk of injury or loss of human life is 1.00×10^{-5} . Without the lightning protection system, the calculated risk value is $R1 = 6.94 \times 10^{-5}$. According to the NFPA Lightning Risk Assessment calculations: **Without a lightning protection system, the expected lightning risk of injury or loss of human life is higher than the tolerable limit.**

According to NFPA 780, the typical tolerable risk of loss of services is 1.00×10^{-3} . Without the lightning protection system, the calculated risk value is $R2 = 6.94 \times 10^{-5}$. According to the NFPA Lightning Risk Assessment calculations: **Without a lightning protection system, the expected lightning risk of loss of services is within the tolerable limit**